What This Cell Does (Overall)

This code checks:
Where your program is running
What files are present
Whether your PDF folder exists

Basically:

"Am I in the correct folder? Are my PDFs available?"

🔍 **Step-by-Step Explanation**

🟢 **1. Get Current Working Directory**

```
cwd = os.getcwd()
print(f" Current Working Directory:\n{cwd}")
```

👉 This tells:

Which folder your notebook/code is currently running in

Example output:

C:\Users\hp\...\rag_2_document

🟢 **2. List Files in Current Folder**

```
for item in os.listdir(cwd):
    print(f"  - {item}")
```

👉 This shows:

All files/folders inside current directory

Example:

- data_docum
- notebook.ipynb
- .env

🟢 **3. Create Path for Data Folder**

```
data_path = os.path.join(cwd, "data_docum")
```

👉 This builds:

Full path to your PDF folder

Example:

C:\Users\hp\...\rag_2_document\data_docum

🟢 **4. Check If Folder Exists**

if os.path.isdir(data_path):

👉 This checks:

Does "data_docum" folder exist or not?

🟢 **5. If Folder Exists**

files = os.listdir(data_path)

👉 This lists:

All PDF files inside data_docum

Example:

- combustion_process.pdf
- engine_system.pdf

🟢 **6. If Folder is Empty**

print("data folder is empty.")

👉 Means:

Folder exists but no files inside

🔴 **7. If Folder NOT Found**

print("data folder NOT found.")

👉 Means:

Your path is wrong OR folder is missing

🧠 Why This Cell is IMPORTANT

Without this:

PDF reading will fail ❌
FileNotFoundError ❌

👉 This cell helps you debug:

✔ Correct folder path
✔ Files present
✔ Setup is correct

🔥 Simple Analogy

Before cooking:
👉 Check kitchen
👉 Check ingredients

This code = checking your "ingredients" (PDFs)

🎯 One-Line Summary

This code verifies the working directory and ensures that the data_docum folder and required PDF files are present before processing them.

🚀 What You Should See

If everything is correct:

✓ Current directory shown
✓ data_docum folder found
✓ PDF files listed

Next cell

🟩 What This Cell Does (Overall)

PDF files → Read → Extract text → Store in dictionary

👉 Output:

```
documents = {
  "file1.pdf": "full text...",
  "file2.pdf": "full text..."
}
```

🔍 **Step-by-Step Explanation**

🟢 **1. PDF File List**

```
pdf_files = [
    "data_docum/combustion_process_60_pages.pdf",
    ...
]
```

👉 This is:

List of all PDF file paths

👉 You are telling Python:

"These are the files I want to read"

🟢 **2. Function Definition**

```
def extract_text_from_pdfs(pdf_paths):
```

👉 You are creating a **function**

Input → list of PDFs
Output → extracted text

🟢 **3. Create Storage**

```
all_texts = {}
```

👉 This is a **dictionary**

Key = file name
Value = full text of that file

Example:

{
"engine.pdf": "Engine consists of..."
}

🟢 **4. Loop Through Each PDF**

for path in pdf_paths:

👉 This means:

Process each file one by one

🟢 **5. Print File Name**

print(f"\n Processing: {path}")

👉 Just for debugging:

Shows which file is being processed

🟢 **6. Check File Exists**

if not os.path.exists(path):

👉 Prevents error:

If file missing → skip it

🟢 **7. Read PDF**

reader = PdfReader(path)

👉 This loads the PDF

🟢 **8. Initialize Empty Text**

text = ""

👉 This will store:

All pages text combined

🟢 9. Loop Through Pages

for i, page in enumerate(reader.pages):

👉 Means:

Read page 1 → page 2 → page 3 ...

🟢 10. Extract Text

page_text = page.extract_text()

👉 Converts:

PDF page → plain text

🟢 11. Append Text

```
if page_text:
    text += page_text + "\n"
```

👉 Combine all pages:

Page1 + Page2 + Page3 → full document text

🟢 12. Store in Dictionary

all_texts[os.path.basename(path)] = text

👉 Example:

```
{
 "engine_system.pdf": "full extracted text"
}
```

👉 basename() removes folder path

🟢 13. Print Success

print(f" Loaded: {file} | Pages: {len(reader.pages)}")

👉 Shows:

File loaded + number of pages

🔴 **14. Error Handling**

except Exception as e:

👉 If something fails:

Print error instead of crashing

🟢 **15. Return Output**

return all_texts

👉 Final output:

Dictionary of all documents

🟢 **16. Call Function**

documents = extract_text_from_pdfs(pdf_files)

👉 Runs the whole process

🟢 **17. Print Output**

print(documents.keys())

👉 Shows:

Which files were loaded

🧠 **Final Data Structure**

```
documents = {
  "combustion.pdf": "text...",
  "engine.pdf": "text...",
  "fuel.pdf": "text...",
  "testing.pdf": "text..."
}
```

🔥 **Why This Cell is VERY IMPORTANT**

Without this:

No text ❌
No triples ❌
No graph ❌
No RAG ❌

👉 This is the **starting point of your pipeline**

🧠 Simple Analogy

PDF = Book
This code = Reading all books and storing notes

🎯 One-Line Summary

This code reads multiple PDF files, extracts their text page by page, and stores the content in a dictionary for further processing in the Graph RAG pipeline.